Blind Noise-Template Subtraction with a Fine-Tuned Demucs Denoiser for Indic Speech Noise Removal

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***Abstract*—Track 2 asks for Vaani speech recordings with their noise events removed, scored by SI-SDR on synthetic clips plus the change in word error rate of a mandated ASR. Our system placed 2nd on the Phase-2 leaderboard with a combined score of 21.24 (SI-SDR 16.77 dB, ΔWER$\times$100 4.47, transcript audit passed). It rests on one observation: the synthetic test clips reuse a small library of noise waveforms. We discover those waveforms blindly from the residual of a denoiser, subtract every exact copy with a least-squares gain, and run a fine-tuned Demucs denoiser on what remains. The denoiser is trained with an SI-SDR loss restricted to event-free target samples. Template subtraction alone added 8.48 points over the denoiser-only system (12.75).**

I. Training Data and Preprocessing

Vaani data. We used the released validation set of Track 2: 695 synthetic clips (noisy mixture, clean reference and noise timestamps) and 1,806 natural noisy clips, all 16 kHz mono. We also used the Gold tier of the Vaani noise-event dataset [1]: 10,772 clips with verified noise-event timestamps. No Phase-2 test labels were used for training or model selection.

Noise bank. A synthetic clip is exactly its clean reference plus pasted noise (outside the pasted spans the residual is about 10^{-4}). The pasted noise of each synthetic validation clip, mixture minus reference, gives 2,213 exact noise snippets. This bank is the only noise source in the final system.

Augmentation pipeline. Training mixtures are made on the fly, mirroring the organisers' generator. (1) A target recording is drawn; of four random 4 s crops the one with the largest event-free share is kept (resampled if below 30%). (2) With probability 0.85, 1–3 bank snippets are pasted at random positions (no noise otherwise). (3) Each snippet is scaled so that its RMS relative to the local target RMS follows a log-normal law centred on the median ratio measured in the validation mixtures ($\sigma = 0.5$ in log scale). (4) Mixture and target are jointly peak-limited to 0.99. Targets are the natural validation clips, the Gold clips and the 695 clean references.

Event-free masks. Natural and Gold recordings almost always contain real noise events (99.9% of clips; mean coverage 27–30%). For every target we build a mask that is 0 inside annotated events dilated by $\pm$150 ms and 1 elsewhere; clean references are fully 1.

Additional datasets. ESC-50 [6] was tested as extra bank noise in one rejected variant (score 12.06 vs. 12.37 without it); it is not used by the final system.

II. Proposed Method

A. Denoiser

The denoiser is Demucs in the dns64 configuration [2] (causal U-Net encoder/decoder with an LSTM bottleneck, hidden size 64, 16 kHz), initialised from the public pretrained dns64 weights [3]. The loss is the negative SI-SDR [4] computed only on masked (event-free) samples m of the target s:

$$\mathcal{L} = -10 \log_{10} \frac{\|\alpha\, m \odot s\|^2}{\|\alpha\, m \odot s - m \odot \hat{s}\|^2}, \tag{1}$$

with $\alpha = \langle m \odot \hat{s},\, m \odot s \rangle / \|m \odot s\|^2$ and $\hat{s}$ the denoiser output. Without the mask the model learns to keep real events, because they are present in its "clean" targets.

Training runs in four stages with AdamW (weight decay 10^{-3}), fp16 mixed precision and gradient clipping at norm 5: (0) 30k steps, batch 8, learning rate 10^{-4} without masks; (1) 30k steps, batch 8, $5{\cdot}10^{-5}$ with event-free masks; (2) 10.3 h on a Tesla T4 (81,482 steps, batch 16, peak $5 \cdot 10^{-5}$, time-based warm-up and cosine decay); (3) 8.5 h continuation at peak $3 \cdot 10^{-5}$ with two seeds (78,220 and 78,631 steps). At inference the two seeds are averaged at the waveform level.

B. Blind noise-template discovery

We measured that pasted noises repeat: 71% of 300 validation noise snippets have a twin with normalised cross-correlation (NCC) $\geq$ 0.95 in another clip, and one horn waveform occurs 69 times. On the test set the noise is unknown, but the denoiser residual $r = x - \hat{s}$ approximates it; 14% of 300 random test clips have a residual twin at NCC ≥ 0.9. Discovery proceeds as follows:

1) *Seeds*: the strongest 1 s window of each clip's residual, strongest first.
2) *Scan*: each seed is correlated with all residuals decimated to 4 kHz (8 CPU workers); hits with NCC ≥ 0.70 are re-aligned at 16 kHz and kept if NCC ≥ 0.75.
3) *Template*: aligned copies ($\pm$1.5 s around the seed) are averaged after least-squares gain normalisation; the template support is where copies agree (coherence ≥ 0.5 on 20 ms frames).
4) *Occurrences*: at each hit the gain g is fitted on the mixture, $g = \langle x, T \rangle / \|T\|^2$.
5) *Pass 2*: steps 1–4 are repeated on $x - \sum gT$ to find noises hidden under the first ones.

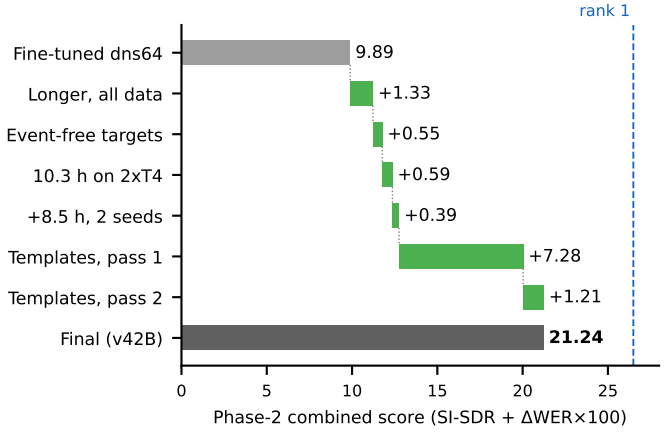


Fig. 1. Phase-2 leaderboard score after each accepted change (green: gain over the previous best).

Pass 1 found 223 templates with 5,234 occurrences in 1,541 test clips; pass 2 added 209 templates and 1,770 occurrences. In total 7,004 occurrences were subtracted in 1,588 clips.

C. Final output and packaging

For clips with occurrences the output is the two-seed denoiser applied to $x - \sum gT$; other clips keep the denoiser-only output. Each waveform is peak-limited to 0.99 and rounded to a multiple of 4 in int16 (quantisation SNR about 68 dB) *before* transcription, so the organisers' audit re-transcribes identical audio. The mandated SraVaani-1.0 ASR [5] transcribes exactly those files, and the zip (deflate level 9) is 924.5 MB, under the 1 GB limit.

D. Pretrained models

- Demucs dns64, facebookresearch/denoiser v0.1.5 [3]: initialisation of the denoiser.
- SraVaani-1.0, ARTPARK-IISc (Hugging Face) [5]: transcripts, as mandated.

III. Results and Conclusion

Fig. 1 shows how each accepted change moved the Phase-2 score. Denoiser improvements together added 2.86 points; template subtraction added 8.48 more. Against the denoiser-only system, the final system is better on 478 of the 500 synthetic clips that Codabench reports per clip, by 4.90 dB SI-SDR on average, with the largest gains on the hardest clips (+6.2 to +6.6 dB where the denoiser alone stayed below 10 dB).

Table I lists the main ablations. Subtraction without the final denoiser scores only 6.12: subtraction removes the repeated noise exactly but leaves residue, unmatched noises and alignment error, which the denoiser removes. The two steps are complementary. Clean references as targets scored below natural targets (9.01 vs. 9.89), because the organisers' references keep the recording's own ambience; the model learned to scrub too much. Extra ESC-50 noise lowered the score on 344 of 500 clips, so the final bank holds only validation noise.

TABLE I
Ablations on the Phase-2 leaderboard

System	Score
Pretrained dns64, no fine-tuning	7.38
Fine-tuned 3k steps, natural validation targets	9.89
same, clean references as targets	9.01
Fine-tuned 30k steps, natural + Gold targets (stage 0)	11.22
+ event-free masked loss (stage 1)	11.77
+ 10.3 h long run (stage 2)	12.37
same, ESC-50 noise added to the bank	12.06
+ 8.5 h continuation, 2-seed mean (stage 3)	12.75
Template subtraction only, no denoiser	6.12
Templates pass 1 + denoiser	20.03
Templates pass 1+2 + denoiser (final)	**21.24**

Conclusion. When a test set is generated from a finite noise library, the noise itself can be recovered without labels by correlating denoiser residuals across clips. Exact subtraction of the recovered waveforms, followed by a learned denoiser, lifted SI-SDR from 11.75 to 16.77 dB and ΔWER from 1.00 to 4.47. Our final score is 21.24, rank 2. The released code reproduces the submitted audio sample for sample.

References

[1] ARTPARK, IISc and Google, "Vaani noise event dataset," Hugging Face, https://huggingface.co/datasets/ARTPARK-IISc/Vaani-Noise-Event-Dataset, 2026.
[2] A. Défossez, G. Synnaeve and Y. Adi, "Real time speech enhancement in the waveform domain," in *Proc. Interspeech*, 2020.
[3] Facebook Research, "denoiser," https://github.com/facebookresearch/denoiser.
[4] J. Le Roux, S. Wisdom, H. Erdogan and J. R. Hershey, "SDR – half-baked or well done?" in *Proc. ICASSP*, 2019.
[5] ARTPARK-IISc, "SraVaani-1.0," https://huggingface.co/ARTPARK-IISc/SraVaani-1.0, 2026.
[6] K. J. Piczak, "ESC: Dataset for environmental sound classification," in *Proc. ACM Multimedia*, 2015.